

1) Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Step 1: Open Command Prompt

- Press Windows + R.
- Type cmd and press Enter.

Step 2: Run the Ping Command

- Type: ping www.youtube.com
- Press Enter and wait for the replies.

Step 3: Record the Result

- Observe the packets sent, received, and lost.
- Note the minimum, maximum, and average response time shown in the output.
- Example result format: Approximate round trip times — Minimum: ____ ms, Maximum: ____ ms, Average: ____ ms.

2) Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Step 1: Open Command Prompt

- Open Command Prompt on your computer.

Step 2: Run Traceroute

- Type: `tracert www.flipkart.com`
- Press Enter and wait for the command to complete.

Step 3: Record the Hops

- List each hop number and the IP address or hostname displayed in the output.
- Each hop represents a network device or routing point through which the traffic passes toward the destination.
- Record the hops exactly as shown on your computer because the route can vary depending on the network and time.

Hop List

- Hop 1: _____
- Hop 2: _____
- Hop 3: _____
- Hop 4: _____
- Hop 5: _____
- Continue the list if additional hops are displayed.

3) Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Step 1: Open Command Prompt

- Open Command Prompt.

Step 2: Run nslookup

- Type: nslookup www.zomato.com
- Press Enter.

Step 3: Record the Result

- Note the DNS server shown in the output.
- Record the IP address or addresses returned for www.zomato.com.
- Example format: Name: www.zomato.com — Address: _____.

4) Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Step 1: Find the Local Device IP Address

- Find the IPv4 address of the other device on the same local network.
- Example: 192.168.1.25.

Step 2: Run the Ping Command

- Open Command Prompt.
- Type: ping 192.168.1.25
- Replace the example address with the actual local device IP address.

Step 3: Observe the Result

- If the device is online and allows ICMP traffic, replies should be received.
- If the device is offline or unreachable, the ping may show Request timed out or a similar failure message.
- Packet loss may occur when the device is unavailable.

Example Observation

- Online device: Reply from the destination IP with response time.
- Offline/unreachable device: Request timed out and the packet loss percentage may increase.

5) Create a short report (5–7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each problem: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Short Report

- Ping is useful for checking whether a destination is reachable and for measuring response time, which helps identify connectivity or latency problems.
- Traceroute is useful when a connection is slow or a route appears to have a problem because it shows the intermediate hops toward the destination.
- Nslookup is useful when a DNS problem is suspected because it checks the DNS resolution of a domain name and displays the returned address information.
- For a website that is not loading, ping can first help determine whether the destination is reachable.
- For a slow connection, traceroute can help identify where delays or routing problems may occur.
- For suspected DNS issues, nslookup is the most directly useful tool for checking domain-name resolution.
- The final diagnosis should be based on the actual outputs obtained during the tests above.

Tool Selection Summary

- Website not loading → Ping (initial reachability check).
- Slow connection → Traceroute (identify route/hop delays).
- Suspected DNS issue → Nslookup (check name resolution).